



**Politecnico
di Torino**

Politecnico di Torino

Lab report for the course Qubit Electronics

Master degree in Quantum Engineering

Group 1

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CHAPTER 1

Intro

A double quantum dot in a generic fully-depleted-on-insulator (FD-SOI) structure is made of three layers. The first layer is made of silicon dioxide, onto which metallic control gates for the quantum dots are deposited. To achieve full double-dot control, three barrier gates and two plunger gates are employed. The barrier gates regulate tunneling between the source, dot 1, dot 2, and the drain, while the plunger gates adjust the detuning of the dot energy levels and their alignment relative to the Fermi levels of the source and drain.

The second layer contains the silicon channel and its surrounding oxide, labeled STI (shallow trench isolation). The channel itself consists of intrinsic silicon, while the source and drain regions are heavily n-doped.

The third layer is a silicon dioxide sheet known as the buried oxide. In this setup, the buried oxide supports quantum confinement of electrons along the vertical direction.

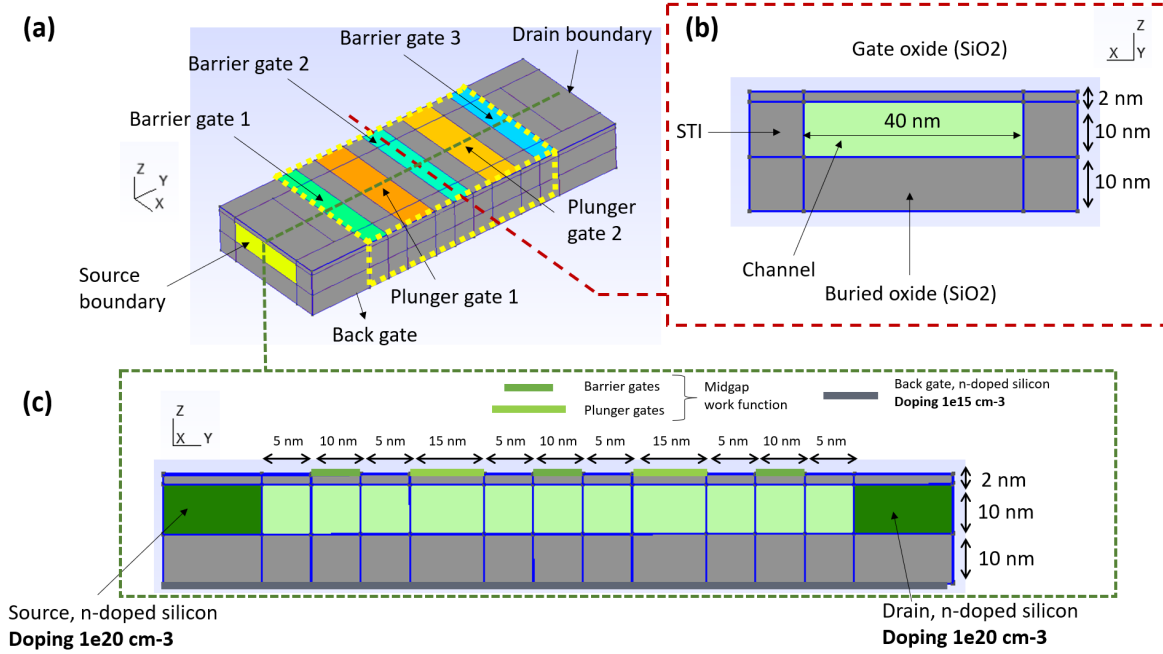


Figure 1.1

CHAPTER 2

Mesh Generation

Question: Explain the process used in the code to initialize the structure and how the dimensions and areas are defined.

In the first part of the code, the variables needed for sizing the structure are defined: the gap length between the various elements, the plunger gate length, the barrier gate length, the source length and the drain length, the domain width (which is the width of the whole structure), the channel width, the gate oxide thickness, the film thickness and the overall thickness. Their dimension is expressed in nm.

In the second part, rectangles are defined to specify the area in x, y plane of the drain, source and simulation domain. The `rectangle(x,y,z,dx,dy,tag)` command is used, where:

- x, y, z are the coordinates of the origin;
- dx and dy the length on the x and y direction respectively.

After that, the rectangles for the plunger and barrier gates are defined, this time in a trickier way.

1. 2 arrays are created, one containing the gap lengths and one containing the length of the plunger and barrier gates.
2. A temporary variable is declared, which corresponds to the half of the length of the channel plus the length of one gap.
3. A for loop is implemented to create alternating plunger and barrier gates. It uses modulo operation ($i\%2$) to alternate the gate lengths.

Thanks to the command `BooleanFragments`, the rectangles are then put together. The next step consists in the upward extrusion of all the surfaces to create the gate oxide layer. The layers created are 2 for computation and mesh purposes.

Physical surfaces for the gate boundaries are defined with the code

```
Physical Surface("<name>") = {n_1, n_2, n_3}.
```

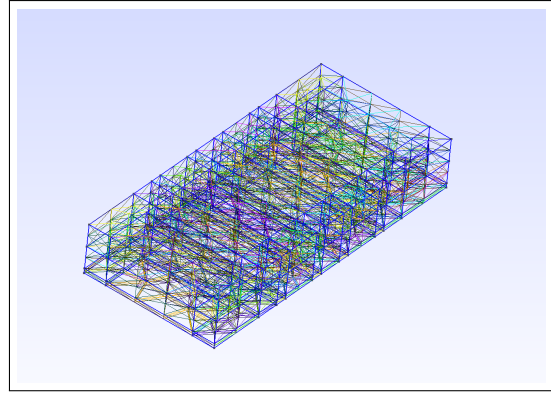
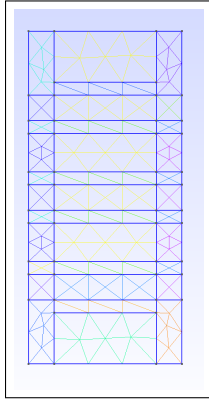
The three numbers in the parenthesis refer to the surfaces used to create the physical one. The same applies for the physical volumes.

Physical groups are uniquely defined by a pair of integers: their dimension (0 for points, 1 for curves, 2 for surfaces, 3 for volumes) and their tag, a strictly positive global identification number. In general, they are used to group elementary geometrical entities into more meaningful groups, e.g. to define some mathematical ("domain", "boundary"), functional or material properties.

The extrusion of the buried oxide is then performed and references for the back gate and the buried oxide are specified.

At the end, the simulation results are saved in two file with two different formats: `.msh2` and `.unrolled`. The first one contains the nodes, which are the coordinates of all mesh points, and the physical groups, i.e. physical surfaces and volumes. This type of file is used as input for solvers. In our case, python. The second one, instead, stores the expanded geometry script after all operations. It useful to check for the right surface or volume IDs (e.g. when inserting the numerical parameters for the physical group) or to rebuild the geometry without running the original script.

Question: Provide a screenshot of the three-dimensional structure in the clearest manner possible



After the execution of the command `gmsh dqdfdsol.geo`, the `.msh2` file and `.unrolled` file have been created automatically.

Question: What are the dimensions of the whole structure along the x, y, and z coordinates? In addition, try to comprehend the base classes and how they are declared. Refer to the manual for further information.

The structure is 60 nm long in the x-direction, 130 nm in the y-direction and 22 nm along z.

A model in Gmsh is defined using its Boundary Representation (BRep): a volume is bounded by a set of surfaces, a surface is bounded by a series of curves, and a curve is bounded by two end points. Model entities are topological entities, i.e., they only deal with adjacencies in the model, and are implemented as a set of abstract topological classes. Examples are *points*, *lines*, *plane surface*, *curve loop*, *surface loop* and *volume loop*.

CHAPTER 3

Schrodinger and Poisson: understanding the script file

Question: Open the file `double_dot_stability.py`. Look at these two lines of code:

```
scaling = 1e-9
mesh = Mesh(scaling, str(path_mesh / "dqdfdsoi.msh2"))
```

What is the function of these two lines of code, what are the parameters inserted for the Mesh class?

The Mesh class accepts as the first parameter the *scaling factor*, which is the factor by which node coordinates are multiplied to correspond to SI units. In our case, this factor is set to represent nm. The second parameter is the path of the `.msh2` file to build the mesh.

Question: Subsequently, look at these two lines of code:

```
dvc = Device(mesh, conf_carriers='e')
dvc.set_temperature(0.1)
```

Explain the function of these lines of code defining what the arguments passed are. Then, explain the line of code related to the temperature, indicating the operating temperature with the appropriate unit of measurement.

The `dvc = Device(mesh, conf_carriers='e')` specifies as a first parameter the mesh on which the device properties are defined and as the second parameter which carrier densities from quantum-confined regions to add to the total charge density. In our case, electrons. The `dvc.set_temperature(0.1)` sets the temperature to 0.1 K.

Question: Identify the materials assigned to each region and compare the names of script files with the defined names in the mesh.

- Oxide: SiO₂;
- Oxide dot: SiO₂;
- Gate oxide: HfO₂;
- Gate oxide dot: SiO₂;
- Buried oxide: SiO₂;

- Buried oxide dot: SiO₂;
- Channel: Si;
- Channel dot: Si;
- Source: n-doped Si;
- Drain: n-doped Si.

The names in the script files coincide with the defined names in the mesh.

Question: By looking at the subsequent lines of code, we can see lines of code like `new_gate_bnd`. Explain the code's arguments, discern within the code the regions of the dot, and the corresponding materials that compound it.

The first argument is the physical surface defined in the `.geo` file. The second argument is the bias, defined earlier in the code and the third argument is the silicon work function.

The dot is defined inside the code by the lines:

```
dvc.new_region("oxide_dot", mt.SiO2)
dvc.new_region("gate_oxide_dot", mt.HfO2)
dvc.new_region("buried_oxide_dot", mt.SiO2)
dvc.new_region("channel_dot", mt.Si)
...
dot_region_list = ["oxide_dot", "gate_oxide_dot", "buried_oxide_dot", "channel_dot"]
dvc.set_dot_region(dot_region_list)
```

Question: Continuing to observe the same code `double_dot_stability.py`, understand when the class is defined, when the methods for solving Poisson and Schrodinger equations are instantiated, and when they are solved. Report the lines.

The class is defined

```
# Define the device object
dvc = Device(mesh, conf_carriers='e')
```

The Poisson and Schrödinger equations are instantiated and solved in these lines.

```
# Instantiate Schrodinger solver's parameters
params_schrod = SchrodingerSolverParams()
params_schrod.num_states = num_states      # Number of states to consider

# Instantiate Poisson solver
poisson_slv = PoissonSolver(dvc, solver_params=params_poisson,
                             geo_file=str(path_mesh/"dqdfdsol.geo_unrolled"))

# Solve Poisson's equation
poisson_slv.solve()

# Solve Schrodinger's equation
schrod_solver.solve()
```

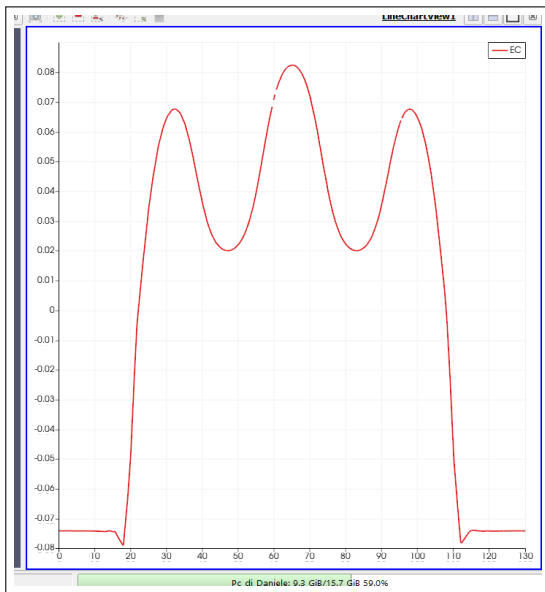
CHAPTER 4

Simulation: Poisson and Schrödinger

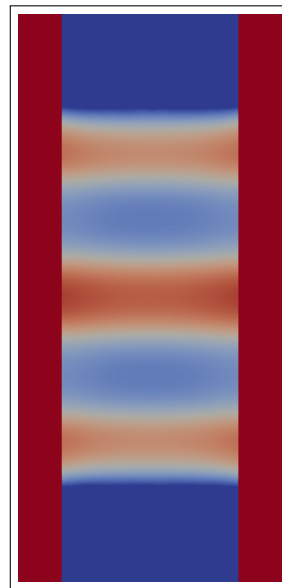
4.1 EC.vtu

Question: Why do we choose to slice at this height and not, for example, at -11? After obtaining the output image, explain what is shown after the slice command, display the image, and describe it briefly also explaining which is the referring color that characterizes the zones that contain a double quantum dot. Describe the plot, show how the potential changes in the domain and what the minima represent; and explain if the behavior in the central region is similar to the expected one or not.

We choose to slice at -2.5 and not at -11, because otherwise we would be too deep in the device and we wouldn't observe the channel.



(a)



(b)

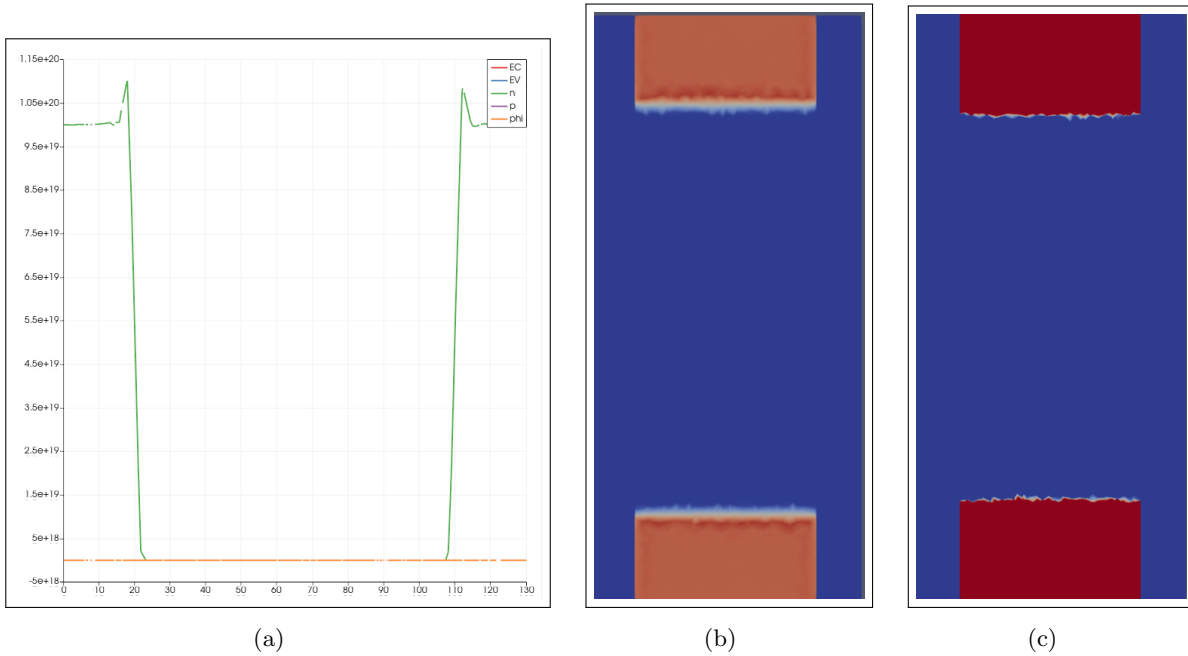
The line plot (a) shows the electrostatic potential across the device, where the two wells corresponding to the dots are clearly visible. The slice (b) shows instead the x-y section of our device. We can clearly distinguish the different regions. The color map shows the spatial distribution of the potential and charge density in the device cross-section. You can see alternating regions of red and blue, representing high (red) and low potential (blue) areas.

- the source and drain regions are in dark blue, with a high presence of electrons due to the n-doping;
- the regions underneath the gate contacts are in light red;
- the dots are in light blue, since only few electrons contribute to the potential.

The behavior at the central point is what we expect from barrier gate 2, biased with a lower potential with respect to the other two barrier gates.

4.2 N.vtu

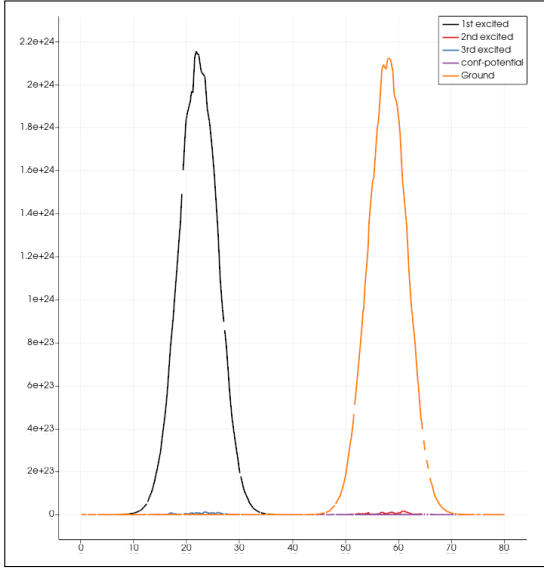
Question: Observing the simulation just obtained, lower the Set Range value, for example, to 10^{18} . Knowing that this represents a simulation obtained as a result of Poisson's solver, explain why the central area of the output image, where the dots are located, appears to be close to zero.



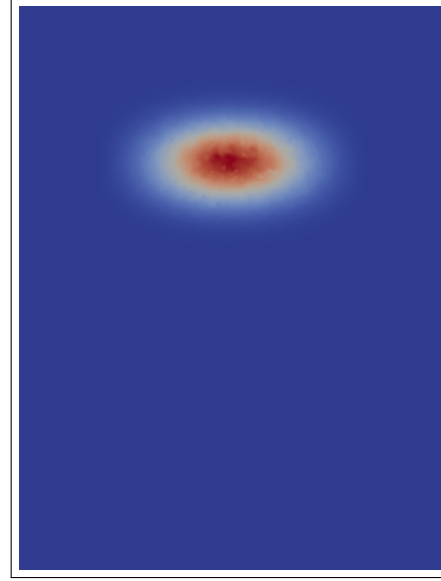
The dots represent discrete charged regions (like dopant atoms, electrons, or holes). However, outside or between the dots, the net charge density is close to zero because the surrounding material is typically neutral. As a result, the Poisson solver outputs a potential value near zero in those neutral regions. When you lower the Set Range, you make the visualization more sensitive to small variations, but the neutral regions will still appear dark blue because the actual computed value is small.

4.3 Q.vtu

Question: Explain what the displayed file represents: what can you notice in the plot? Explain how the curve behaves in both the ground state and the first excited state, compare the two curves, and indicate their respective configurations



(a) 1D plot of the probability density along y-axis



(b) 2D color map of the probability density

Image (a) shows the probability density along the length of the device: the orange line corresponds to the ground state, while the black line represents the first excited state. Each curve displays a single peak at a different position, indicating that in both states the electron is localized in a specific region. These peaks effectively reveal the locations of the quantum dots.

Image (b) shows a spatial map of the probability density for finding the electron in the ground state in our device. Red indicates high probability, while blue indicates low probability. The map shows a single, localized region of high probability, consistent with the ground state localization observed in (a). Based on the line plot, we can conclude that the second dot most likely hosts the electron in its ground state.